

Sustainable Food Production

Science

May, 2026



Sustainable Food Production (A11485)

Short Title: Sustainable Food Prod.
Faculty Science and Computing
Department Science
Cluster Horticulture
Author CPWALSH
Credits 10

Level: Intermediate

Description of Module / Aims

The module will enable the student to plan and describe the production of a range of speciality food crops and perform the associated skills to best industry practices.

Programmes

F00D-0023	BSc in Horticulture (WD_SHORT_D)	2 / 3 / E
F00D-0023	BSc in Horticulture (WD_SHORT_D)	3 / 5 / E
F00D-0023	BSc in Horticulture (WD_SHORT_DX)	2 / 3 / E
F00D-0023	BSc in Horticulture (WD_SHORT_DX)	3 / 5 / E

Indicative Content

- Salad crops, scallions, lettuce and celery: Soil, site and rotation. Cultivars. Sowing/planting and irrigation. Cultivation and nutrition. Weed, pest and disease control including SUD/IPM. Harvesting and grading
- Rhubarbs, asparagus, courgettes and sweetcorn: Site selection and soils. Cultivars. Cultivation and nutrition. Weed, pest and disease control including SUD/IPM. Harvesting and grading
- Leeks, peas and beans: Site selection and soils. Cultivars. Cultivation and nutrition. Weed, pest and disease control including SUD/IPM. Harvesting and grading
- Beetroot, radish and spinach: Site selection and soils. Cultivars. Cultivation and nutrition. Weed, pest and disease control including SUD/IPM. Harvesting and grading
- Herbs (culinary and medicinal): Market size and outlet. Range of herb crops. Site selection and soils. Cultivation and nutrition. Weed, pest and disease control including SUD/IPM. Harvesting and grading
- Blueberries, blackberries and currants: Propagation methods. Choice of varieties. Nutrition Soil preparation and planting. Pest disease and weed control including SUD/IPM. Picking, training and pruning
- Nitrates Directive, FSAI regulations, Bord Bia QA scheme, Origin Green, Pesticide Control/Registration
- Organic Conversion: Time periods, and preparation of application form
- Pest and disease management in an organic system 'v' conventional systems
- Soil fertility and management in an organic system 'v' conventional systems
- Regulation of organic production, grading and marketing of produce

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse the propagation and production of a range of food crops (including vegetables, fruit and herbs) under sustainable systems.
2. Produce a conversion plan for an organic production system.
3. Complete routine tasks associated with specialty crops.
4. Relate the requirements of the quality assurance schemes, EU directives, and other regulations to management strategies in food production systems.

Learning and Teaching Methods

- Lecture
- Practical demonstration
- Project
- Practice

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,4
Continuous Assessment	50%	
<i>Project</i>	25%	2
<i>Practical</i>	25%	3

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrates a limited knowledge of the subject matter. May have some problems with critical thinking and problem solving. Adequate practical skills.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Good practical tasks.

60%-69%: Demonstrates skilled and sound knowledge of the subject matter. Shows ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.

70%-100%: All of the above to an excellent level. Demonstrates authoritative handling of complex material and a highly developed and extensive understanding of the subject.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Practical	48	
Independent Learning	174	

Essential Material

- Pollock, M. *RHS Vegetable & Fruit Gardening*. London: Dorling Kindersley, 2012.

Supplementary Material

- "Protected Crop Growing, A guide for home and Market Gardeners." <http://www.teagasc.ie/>. <http://www.teagasc.ie/publications/2011/1000/ProtectedCropGrowing.pdf>
- Dixon, G.R. and D.E. Aldous. *Horticulture: Plants for People and Places, Volume 1*. Amsterdam: Springer Netherlands, 2014.
- Larkcom, J. *The Organic Salad Garden*. London: Frances Lincoln, 2006.

Requested Resources

- Lecture Room: Loose Seated